Anis ur Rehman
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https://scholar.google.com.au/citations?hl=en&user=l1f8c9QAAAAJ&view_op=list_works&sortby=pubdate

BASIC INFORMATION

First Name: Anis ur **Sur Name:** Rehman

Email: engr.anisurrehman861@gmail.com
ORCID ID: 0000-0002-1878-4523
LinkedIn ID: https://www.linkedin.com/in/engr-anis-ur-rehman-a8a135166
Google Scholar: https://scholar.google.com/citations?user=l1f8c9QAAAAJ&hl=en
Race for 2030: https://www.racefor2030.com.au/team/anis-ur-rehman/

EDUCATION

Griffith University **Queensland, Australia**
Ph.D in Electrical Engineering Jul, 2024 – Continue
- Project/Thesis: Demand management of V2G and V2H using microgrid and energy storage-based EV charging system
- Current Status: Passed confirmation

University of Azad Jammu and Kashmir **Azad Kashmir, Pakistan**
M.S Electrical Engineering Nov, 2020 – Nov, 2022
- Thesis: Multi-agent distributed reinforcement learning framework based distributional voltage control with high penetration of renewable energy resources
- Concentration: Machine learning
- CGPA: 3.93/4.0

Mirpur University of Science and Technology **Azad Kashmir, Pakistan**
B.S Electrical Engineering (with specialization in Power Engineering) Oct, 2015 – Oct, 2019
- Thesis: Feasibility Study for the integration of renewable energy resources
- CGPA: 3.43/4.0

RESEARCH INTEREST

- Electrical Vehicles
- Renewable Energy
- Artificial Intelligence
- Reinforcement Learning
- Energy Management
- Charging Stations Optimization

TECHNICAL SKILLS

- **Programming Languages**: Python, Machine Learning Libraries, Neural Networks, Pytorch, Tensorflow
- **Power System Software**: HOMER, MATLAB, DIgSILENT Power Factory, OpenDSS, Arduino IDE
- **Other Software**: Microsoft Office, Origin, Adobe Photoshop, Adobe Illustrator,

PUBLICATIONS

[1] **Anis ur Rehman,** J. Lu, B. Du, F. Bai, M. A. Hossain, and M. J. Sanjari, "Systematic review of reward function design in reinforcement learning for EV charging and battery energy storage," *Renew. Sustain. Energy Rev.*, vol. 243, Art. no. 117366, 2027, doi: 10.1016/j.rser.2026.117366.

[2] **Anis ur Rehman**, P. Akhgar, M. Soederberg, and M. J. Sanjari, "Why stay connected? Understanding customer transitions in the modern energy landscape," *Renew. Sustain. Energy Rev.*, vol. 241, Art. no. 117270, 2026, doi: 10.1016/j.rser.2026.117270**.**

[3] **Anis ur Rehman**, H. M. Khalid, A. Ismail, R. Alhejaili, A. Gulraiz, Z. Said, and G. A. Hussain, "An IoT integrated home energy management system: An optimized solution towards real-time energy resources monitoring and climate-based decision support," International Journal of Electrical Power & Energy Systems, vol. 177, p. 111836, 2026, doi: 10.1016/j.ijepes.2026.111836.

[4] **Anis ur Rehman**, "Reinforcement learning as a control layer for electric vehicle interaction with multi-energy systems: A comprehensive review," Renew. Sustain. Energy Rev., vol. 231, p. 116733, 2026, doi: 10.1016/j.rser.2026.116733.

[5] **Anis ur Rehman,** M. J. Sanjari, R. M. Elavarasan, and T. Jamal, "Sustainability-aligned pathways for energy transition: A review of low-carbon energy network solutions," Renewable and Sustainable Energy Reviews, vol. 226, pp. 116428, 2026.

[6] **Anis ur Rehman**, J. Lu, B. Du, F. Bai, M. J. Sanjari, and M. A. Hossain, "Vehicle-to-grid technology for load balancing and energy management: A comprehensive review of technical, economic and environmental perspectives," Applied Energy, vol. 402, pp. 126974, 2026.

[7] **Anis ur Rehman**, J. Lu, B. Du, F. Bai, and M. J. Sanjari, "Efficient Management of Electric Vehicle Charging Stations: Balancing user preferences and grid demands with energy storage systems and renewable energy," Applied Energy, vol. 393, pp. 126147, 2025.

[8] **Anis ur Rehman,** I. G. M. Alblushi, H. M. Khalid, Z. Said, A. Ismail, S. M. Muyeen, H. F. Oztop, and A. Y. Abdelaziz, "Analysis of renewable energy resources (RERs) under the China-Pakistan economic corridor (CPEC) initiative: A proposed solution for optimized and sustainable energy management," Sustainable Futures, vol. 10, no. 100826, pp. 1–X, Jun. 2025.

[9] **Anis ur Rehman,** M. F. Zia, H. M. Khalid, Z. Said, and S. M. Muyeen, "Wind and grid energy-based onshore beach charging station for electric vehicles: An integration infrastructure with techno economics, sustainable mobility, and environmental protection," Results in Engineering, vol. 26, pp. 105113, 2025.

[10] **Anis ur Rehman,** I. G. M. Alblushi, H. M. Khalid, Z. Said, A. Iqbal, and S. M. Muyeen, "Techno economics and energy dynamics of a solar powered smart charging infrastructure for electric vehicles with advanced IoT based monitoring and RFID based security," Sustainable Futures, vol. 8, pp. 100376, 2024.

[11] **Anis ur Rehman,** Z. Ullah, H. S. Qazi, H. M. Hasanien, and H. M. Khalid, "Reinforcement learning-driven proximal policy optimization-based voltage control for PV and WT integrated power system," Renewable Energy, vol. 227, p. 120590, Jun. 2024, doi: 10.1016/j.renene.2024.120590.

[12] **Anis ur Rehman,** Z. Ullah, A. Shafiq, H. M. Hasanien, P. Luo, and F. Badshah, "Load management, energy economics, and environmental protection nexus considering PV-based EV charging stations," Energy, p. 128332, Jul. 2023, doi: 10.1016/j.energy.2023.128332.

[13] **Anis ur Rehman**, A. Shafiq, Z. Ullah, S. Iqbal, and H. M. Hasanien, "Implications of smart grid and customer involvement in energy management and economics," Energy, p. 127626, Apr. 2023, doi: 10.1016/j.energy.2023.127626.

[14] **Anis ur Rehman**, S. Iqbal, A. Shafiq, N. F. Alshammari, S. H. Dar, and R. T. Iqbal, "Multifaceted impacts of widespread renewable energy integration on socio-economic, ecological, and regional development," Sustainable Futures, vol. 8, pp. 100241, 2024.

[15] **Anis ur Rehman**, M. Ali, S. Iqbal, A. Shafiq, N. Ullah, and S. A. Otaibi, "Artificial Intelligence-Based Control and Coordination of Multiple PV Inverters for Reactive Power/Voltage Control of Power Distribution Networks," Energies, vol. 15, no. 17, p. 6297, Aug. 2022, doi: 10.3390/en15176297.

[16] **Anis ur Rehman** et al., "Transition towards a sustainable power system: MA-DA&DC framework based voltage control in high PV penetration networks," Energy Reports, vol. 9, pp. 5922–5936, Dec. 2023, doi: 10.1016/j.egyr.2023.05.035.

[17] **Anis ur Rehman**, H. M. Khalid, and S. M. Muyeen, "Grid-integrated solutions for sustainable EV charging: a comparative study of renewable energy and battery storage systems," *Frontiers in Energy Research*, vol. 12, 2024.

[18] **Anis ur Rehman**, Y. Alamoudi, H. M. Khalid, A. Morchid, S. M. Muyeen, and A. Y. Abdelaziz, "Smart agriculture technology: An integrated framework of renewable energy resources, IoT-based energy management, and precision robotics," Cleaner Energy Systems, vol. 9, pp. 100132, 2024.

[19] **Anis ur Rehman,** I. G. M. Alblushi, M. F. Zia, H. M. Khalid, U. Inayat, M. Benbouzid, S. M. Muyeen, and G. A. Hussain, "A solar-powered multi-functional portable charging device (SPMFPCD) with internet-of-things (IoT)-based real-time monitoring—An innovative scheme towards energy access and management," Green Technologies and Sustainability, vol. 3, no. 1, pp. 100134, 2025.

[20] **Anis ur Rehman,** "Economic and strategic challenges in microgrid integration: Insights from operational dynamics and renewable energy potential," Green Technologies and Sustainability, vol. 3, no. 1, pp. 100130, 2025.

[21] Z. Ullah, **Anis ur Rehman** et al., "IoT-based monitoring and control of substations and smart grids with renewables and electric vehicles integration," Energy, p. 128924, Aug. 2023, doi: 10.1016/j.energy.2023.128924.

[22] A. Bukhari, O. I. Aboulola, **Anis ur Rehman**, A. Alharbi, W. Alosaimi, and A. Daud, "Renewable energy driven on-road wireless charging infrastructure for electric vehicles in smart cities: A prototype design and analysis," Energy Reports, vol. 12, pp. 5145–5154, 2024.

[23] Z. Ullah, H. S. Qazi, **Anis ur Rehman**, H. M. Hasanien, S. Wang, M. R. Elkadeem, and F. Badshah, "Efficient energy management of domestic loads with electric vehicles by optimal scheduling of solar-powered battery energy storage system," Electric Power Systems Research, vol. 234, pp. 110570, 2024.

[24] Z. Ullah, L. Yan, **Anis ur Rehman**, H. S. Qazi, W. Xiaodong, J. Li, and H. M. Hasanien, "Distributed consensus-based optimal power sharing between grid and EV charging stations using derivative-free charging scheduling," *IEEE Access*, vol. 12, pp. 127768–127781, 2024.

[25] Sheeraz Iqbal, Mishkat Ullah Jan, **Anis ur Rehman** et al., "Feasibility Study and Deployment of Solar Photovoltaic System to Enhance Energy Economics of King Abdullah Campus, University of Azad Jammu and Kashmir Muzaffarabad, AJK Pakistan," in IEEE Access, vol. 10, pp. 5440-5455, 2022, doi: 10.1109/ACCESS.2022.3140723.

[26] Aqib Shafiq, Sheeraz Iqbal, **Anis ur Rehman** et al., "Integration of Solar Based Charging Station in Power Distribution Network and Charging Scheduling of EVs," Frontiers in Energy Research, Jan. 2023, doi: 10.3389/fenrg.2023.1086793/abstract.

[27] Naveed, A., Iqbal, S., Munir, S., **Anis ur Rehman**, Eslami, M., Kamel, S.: Renewable energy integration in healthcare systems: A case study of a hospital in Azad Jammu and Kashmir. IET Renew. Power Gener. 18, 796–809 (2024). https://doi.org/10.1049/rpg2.12946.

[28] Zia Ullah, Shaorong Wang, Guan Wu, Hany M. Hasanien, **Anis ur Rehman** et al, " Optimal scheduling and techno-economic analysis of electric vehicles by implementing solar-based grid-tied charging station", Energy, vol. 267, p. 126560, 2023, doi: 10.1016/j.energy.2022.126560.

[29] Sheeraz Iqbal, Salman Habib, Muhammad Ali, Aqib Shafiq, **Anis ur Rehman** et al., "The Impact of V2G Charging/Discharging Strategy on the Microgrid Environment Considering Stochastic Methods," Sustainability, vol. 14, no. 20, p. 13211, Oct. 2022, doi: 10.3390/su142013211.

[30] Aqib Shafiq, Sheeraz Iqbal, Salman Habib, Atiq ur Rehman, **Anis ur Rehman** et al., "Solar PV-Based Electric Vehicle Charging Station for Security Bikes: A Techno-Economic and Environmental Analysis," Sustainability, vol. 14, no. 21, p. 13767, Oct. 2022, doi: 10.3390/su142113767.

[31] Obaid U. Rehman Abbasi, Syed Basit Ali Bukhari, Sheeraz Iqbal, Salik Wasim Abbasi, **Anis ur Rehman**, Energy management strategy based on renewables and battery energy storage system with IoT enabled energy monitoring. Electr Eng (2023). https://doi.org/10.1007/s00202-023-02133-6.

[32] Muhammad Zeshan Afzal, Muhammad Aurangzeb, Sheeraz Iqbal, Mukesh Pushkarna, **Anis ur Rehman** et al., "A Novel Electric Vehicle Battery Management System Using an Artificial Neural Network-Based Adaptive Droop Control Theory," International Journal of Energy Research, vol. 2023, p. 2581729, Jul. 2023, doi: 10.1155/2023/2581729.

Conference Papers

[33] **Anis ur Rehman et al.,** "Control and Coordination of Multiple PV Inverters in Power Distribution Network using Multi-Agent Deep Reinforcement Learning," 2022 International Conference on Emerging Technologies in Electronics, Computing and Communication (ICETECC), Jamshoro, Sindh, Pakistan, 2022, pp. 1-7, doi: 10.1109/ICETECC56662.2022.10069331.

[34] **Anis ur Rehman**, M. J. Sanjari, B. Du and J. Lu, "Potential of PV and Wind Energy-Based EV Charging Stations to Minimize Peak Load Demand," 2025 4th International Conference on Smart Grid and Green Energy (ICSGGE), Sydney, Australia, 2025, pp. 443-448, doi: 10.1109/ICSGGE64667.2025.10985236.

[35] **Anis ur Rehman**, "Hardware-Embedded Digital Twins for Advanced Monitoring and Management in Cyber-Physical Smart Grids with RERs and BESS Integration," 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC), Jamshoro, Pakistan, 2025, pp. 1-6, doi: 10.1109/ICETECC65365.2025.11071254.

[36] **Anis ur Rehman**, M. J. Sanjari, J. Lu and B. Du, "Reinforcement Learning for Battery-Integrated Energy Systems Control: A Comprehensive Review," 2025 IEEE PES 35th Australasian Universities Power Engineering Conference (AUPEC), Brisbane, Australia, 2025, pp. 1-5, doi: 10.1109/AUPEC66173.2025.11219647.

[37] **Anis ur Rehman**, M. J. Sanjari, B. Du and J. Lu, "Internet of Things-Based Accident Detection and Hazard Response System for Intelligent Transportation," 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC), Gold Coast, Australia, 2025, pp. 1885-1890, doi: 10.1109/ITSC60802.2025.11423823.

[38] Aqib Shafiq, Sheraz Iqbal, Syed Danish Ali, **Anis ur Rehman,** "Economic and environmental analysis for different scenarios of grid-connected Solar PV-based EV charging Station facility using Homer Grid," 2022 International Conference on Emerging Technologies in Electronics, Computing and Communication (ICETECC), Jamshoro, Sindh, Pakistan, 2022, pp. 1-5, doi: 10.1109/ICETECC56662.2022.10069720.

WORK EXPERIENCE

Research/Publication

M.S Electrical Engineering

- Machine Learning, Reinforcement learning,
- Electrical Vehicles, Energy Management, Energy Optimization
- Python, Tensorflow, Numpy, Pandas, Keras, Pytorch

Qadirabad Power Station, Power Development Organization (PDO) **Azad Kashmir, Pakistan**

Internee Engineer Sep, 2018- Nov, 2018

- Dam and Run of River Schemes
- Power Transformer and Auto-Transformer
- Governor and Excitation System
- Protection Relay and Protection System of Hydel Power Plants
- Hydraulic Turbine and Generator
- Operation and Maintenance of Hydel Power Plants

Janan & Sons SMC-PVT LTD **Azad Kashmir, Pakistan**

Assistant Engineer 01 Jan, 2021 - 28 Feb, 2023

- Develops an in-depth knowledge of the contractual requirements for the project.
- Oversees the construction to ensure adherence to approved drawings and specifications; inspects, construction workmanship, materials, and equipment.
- Oversees site work elements within the discipline and provides early notification of potential schedule slippages regarding electrical construction activities.
- Monitors the work process, performs quality assurance testing and enforces safety requirements.
- Makes cost estimates on design changes, and evaluates cost proposals and contractors' requests for adjustments.
- Reports to the Project Director on the project's conformity and non-conformity to approved drawings and specifications; make recommendations on how to correct deficiencies and resolve problems during construction.

Robootika, National Incubation Centre **Azad Kashmir, Pakistan**

Co-Founder 01 Mar, 2023 – 01 June 2024

- Robotics hardware designing and simulation
- Coding and troubleshooting
- STEM education trainer

Huazhong University of Science and Technology **China**
Visiting Researcher (Remote) Dec, 2022 – Dec, 2023

- International research collaboration with Dr. Zia Ullah in State Key Laboratory of Advanced Electromagnetic Engineering and Technology, Huazhong University of Science and Technology, China.

CERTIFICATIONS AND ENDORSEMENTS

- HOMER Pro Software for Sizing and Economic Analysis of Hybrid Power Systems
- Basics of Solar PV System and Design using Helioscope
- Elevate Your Leadership Skills to Take Your Organization to the Next Level
- Open channel design for Hydropower and Irrigation
- IC Manufacturing Process for Electronic Systems
- Industrial Wastewater Treatment and Recycling Technologies
- Introduction to IoT and Robotics

HARDWARE-BASED PROJECTS

- Water Monitoring of the Garden and Controlling through Android Application.
- Remote Current and Voltage Monitoring of the substation using Mobile application.
- Robotic Car Follows the Specific Path.
- Smart Robotic Avoid the Obstacle.
- Robot Controlled by the Mobile Application.

Declaration:
I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. All supporting documents and certificates will be provided upon request.